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20 December 1966
SUPERSEDING
MIL-R-7051
30 October 1950

MILITARY SPECIFICATION

RELAY BANK, BOMB SALVO CIRCUIT 3 UNIT, 28 VOLT DC, TYPE A-1

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of relay bank for bomb salvo circuit designated Type A-1. The Type A-1 salvo control relay bank covered by this specification is intended for use in all "Electric Bomb" release systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-250/8	Aluminum Alloy, 5052, Plate and Sheet
QQ-S-571	Solder; Soft (Tin, Tin-Lead, and Lead-Silver

MILITARY

MIL-M-14	Molding Plastics and Molded Plastic Parts, Thermosetting
MIL-P-997	Plastic-Material, Laminated, Thermosetting, Electrical-Insulating: Sheets, Glass Cloth Silicone Resin
MIL-E-5272	Environmental Testing; Aeronautical and Associated Equipment, General Specification for
MIL-C-6055	Drums, Metal Reusable, Shipping and Storage Cap. From 88 to 510 Cubic Inches
MIL-S-6872	Soldering Process, General Specification for
MIL-F-7179	Finishes and Coatings: General Specification for Protection of Aerospace Weapons Systems, Structures and Parts
MIL-Q-9858	Quality Program Requirements
MIL-D-1000	Drawing, Engineering and Associated List
MIL-P-15037	Plastic-Material, Laminated Thermosetting, Sheets, Glass Cloth, Melamine-Resin

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MIL-S-22473	Sealing, Locking, and Retaining Compounds Single-Component
MIL-C-45662	Calibration of Standards

STANDARDS

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Specifications and Standards, Use of
MIL-STD-704	Electrical Power, Aircraft, Characteristics of
MS3102	Connector, Receptacle Electric Box Mounting
MS33586	Metals, Definition of Dissimilar

3. REQUIREMENTS

3.1 General - The individual part requirements shall be as specified herein.

3.2 Qualification - Relay banks furnished under this specification shall be a product which has been tested and passed the qualification tests specified herein, and has been approved. In addition, the supplier shall obtain certification from the responsible government inspection service that he has met the quality control requirements of MIL-Q-9858 as specified herein (see 4.1.3)

3.3 Materials - The materials shall conform to requirements specified herein. When a definite material is not specified, the selection of material shall be at the discretion of the manufacturer. Whenever practicable, the manufacturer shall select materials standards and specification in accordance with MIL-STD-143. Materials selected shall be such that the relays will meet the performance requirements and product characteristics specified herein. The materials shall be of the best quality and lightest practicable weight for the intended purposes. After qualification, any change of parts or materials shall be submitted to the government qualifying activity for approval. Acceptance or approval of any consistent part or material shall not be construed as a guarantee of acceptance of the finished product.

3.3.1 Metals - Metals shall be of a corrosion-resistance type or shall be suitably plated to resist corrosion. The use of dissimilar metals in contacts which tend toward electrolytic corrosion (particularly brass, copper, or steel in contact with aluminum or aluminum alloy) will not be acceptable; however, metal plating or metal spraying of dissimilar base metals to provide similar abutting surfaces is permitted. The use of dissimilar metals separated by suitable insulating material is also permitted. Dissimilar metals are defined in MS33586.

3.3.2 Non-Metals - For external parts of sealed components and all parts of unsealed components, non-metals, including plastics, fabrics and protective finishes, shall be moisture and flame resistant, shall not support fungus growth, and shall not be adversely affected by weathering and aircraft fluids at the temperatures specified herein.

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3.3.2.1 Plastic Material - Except as specified herein, plastic material shall conform to specification MIL-M-14 for molded material and specification MIL-P-997, or MIL-P-15037, whichever is applicable. In addition to these specifications and the requirements of 3.3.2 the plastic material shall not support combustion, give off noxious gases in harmful quantities, give off any gases in quantities sufficient to cause explosion of sealed housings, give off any gases in a sealed housing that will cause contamination of the contacts or other parts of the components or form current-carrying tracks when subjected to arcing conditions encountered when any of the tests in this specification are performed.

3.3.3 Selection of Materials - Specifications and standards for all materials, parts, and government certification and approval or processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided in the following paragraph.

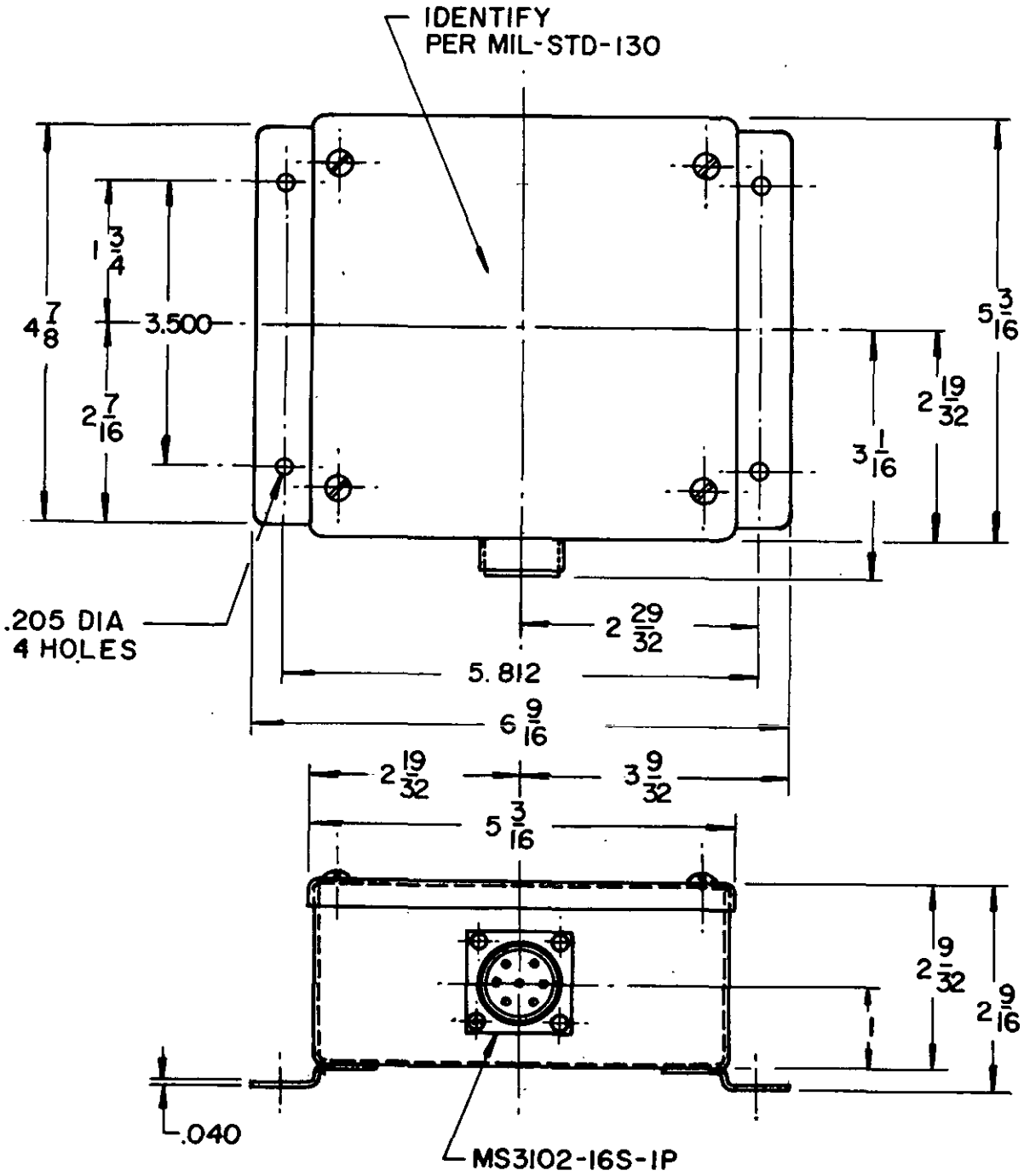
3.3.3.1 Standard Parts - Standard parts (MS, AN, and JAN) shall be used whenever they are suitable for the purpose and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, etc., may be used provided they possess suitable properties and are replaceable by the standard parts (MS, AN, JAN) without alteration. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.4 Design and Construction - The relay bank shall consist of three relays mounted in a sealed metal box to function as a cycling electrical ratchet in the salvo release circuit of bombardment aircraft. The first incoming impulse, from the salvo initiating circuit, shall close and lock in the output relay and output contacts. The second impulse, which may originate either from the door closing or the salvo circuit shall unlock the output relay and open its output contacts, restoring the component relays to their original condition ready for additional recycling operation. The relay units shall be of compact design and of sufficiently rugged construction to withstand the mechanical shocks and stresses incident to their use. The relay units shall be secured to the assembly by mechanical means. The relay parts shall be secured to the assembly by mechanical means. The dimensions of the relay bank shall be as specified on Figure 1. The metal box in which the relays are mounted shall be constructed of aluminum alloy in accordance with QQ-A-250/8.

3.4.1 Electrical Power - The equipment shall meet the applicable requirements of MIL-STD-704 and shall give specified performance when operating from applicable power source as defined therein.

3.4.1.1 Operating Voltage Range - Operating voltage range shall be from 18 to 30 volts for nominal 28 volt equipment. Minimum and maximum operating voltages shall be 18 and 30 volts DC respectively except when otherwise specified.

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UNLESS OTHERWISE SPECIFIED
TOLERANCES ON:-
FRACTIONS - $\pm \frac{1}{64}$
DECIMALS - ± .010

FIGURE 1 Relay Configuration

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3.4.2 Electrical Components - The electrical components parts shall be in accordance with MIL-D-9402 and as specified herein:

a. Relay X - Output Relay:

Coils - opposing - double wound - two section, current drain per coil 0.225 amp. max at 28 volts.

Voltage - 18 to 30 volts DC inclusive.

Make - slow "make" operation, not less than 0.06 of a second.

Break - "break" operation not more than 0.025 of a second.

Contact Rating - 3 amperes minimum.

Contact Arrangement - Contacts X2 and X3 "make" before contacts at X1 "break"; all other contacts "make" and "break" simultaneously.

Operation - (See para 3.4.2.2)

Mounting - supporting bracket or base to be drilled and tapped for 2 No. 8-32 screws.

b. Relay Y - Cycling Relay:

Coils - opposing - double wound - two section, current drain per coil 0.225 amperes maximum at 28 volts.

Voltage - 18 to 30 volts DC inclusive.

Make - "make" operation not more than 0.025 of a second.

Break - "break" operation not more than 0.025 of a second.

Contact Rating - 3 amperes minimum.

Contact Arrangement - contacts Y1 and Y3 "make" and "break" simultaneously with contacts Y4 and Y5. Contacts Y1 and Y2 "break" before contacts Y1 and Y3 "make"

Operation - (See para 3.4.2.2)

Mounting - supporting bracket or base, to be drilled and tapped for 2 No. 8-32 screws.

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c. Relay Z - Door Closing Circuit Relay:

Coils - single winding - current drain not more than 0.04 amperes at 28 volts.

Voltage - 18 to 30 volts DC inclusive.

Make - "make" operation not more than 0.025 of a second.

Break - "break" not more than 0.025 of a second.

Contact Rating - 3 amperes minimum.

Contact Arrangement - normally open

Operation - (See para 3.4.2.2)

Mounting - supporting bracket or base, to be drilled and tapped for 4 No. 3-48 screws.

3.4.2.1 Electrical Connections - The electrical connections to the relay bank shall be made by means of a standard connector receptacle MS3102-16S-1P which shall be securely fastened to the case by 4 No. 4-40 machine screws, lock washers, and nuts.

3.4.2.2 Operation Requirements - The first momentary impulse, (not less than 0.10 of a second) applied to terminal D shall accomplish the following:

a. Energize coil C through contacts Y1 and Y2 to ground.

b. Energize coils A and D simultaneously through contacts at X1 through coils A and D to ground.

c. Actuates relay X due to only coil A being energized, "locks in" through contacts X2 and X3, and breaks the contacts at X1.

d. Relay Y does not actuate due to the opposing fields existing whenever both coils C and D are energized.

e. Positive potential is applied through terminal F to -

(1) Terminal C through contacts X4 and X5.

(2) Terminal B through contacts X6 and X7.

(3) Terminal A through contacts X8 and X9.

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When the momentary impulse is terminated, coil C becomes de-energized; Coil D remains energized by the circuit through contacts X3 and X2; causing relay Y to actuate and thus establish two provisional circuits; provisional circuit P-1, from terminal D through Y1, Y3, coil B to ground; Provisional circuit P-2, from Z2 through Y4, Y5, Coil B to ground. A second momentary impulse applied to terminal D shall pass through provisional circuit P-1 (D through Y1, Y3, Coil B to ground), causing relay X to drop-out because of the opposing fields existing when both coils A and B are energized. At the termination of the second momentary impulse at terminal D, relays X and Y will be restored to their original condition, ready for additional recycling operation. Instead of the second impulse being applied at Terminal D, it is applied at terminal E, relay Z shall actuate and current shall pass from Z1 through the provisional circuit P-2. This shall energize coil B and because Coil A is already energized, the opposing fields existing when these two coils are both energized shall cause relay X to drop-out, returning relays to their original condition ready for additional recycling operation.

3.4.3 Dielectric Strength - The relay units shall be designed to withstand the test voltages specified in 4.5.2.1.

3.4.4 Attitude - The relay bank shall be constructed to insure proper operation when mounted in any position.

3.4.5 Mounting Bracket - When brackets are used, they shall be an integral part of the housing or shall be securely attached thereto in such a manner to prevent any movement between the housing and the mounting bracket in service use.

3.4.6 Enclosures - Enclosures shall be of sufficient mechanical strength to withstand the normal abuse incurred in handling, transit, storage and installation without causing malfunction or distortion of parts.

3.4.6.1 Sealing - The relay bank shall be totally enclosed for mechanical and dust protection and explosion-proofing. The relay bank shall be sealed prior to delivery (see 4.5.1.2).

3.4.7 Finish - Finish of relay bank box shall meet the requirements of MIL-F-7179, FED STD 595, No. 34086.

3.4.8 Screw Threads - Screw threads holding the coils of relay units to their base shall be secured with a retaining compound conforming to specification MIL-S-22473.

3.4.9 Size and Weight - Size and weight shall not exceed the values specified on Figure 1.

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3.4.10 Solder - Whenever suitable for the purpose, compositions SN70, SN60 and SB5, conforming to specification QQ-S-571 shall be used, Solder of other compositions may be used, provided they possess superior characteristics and meet the requirements of this specification.

3.4.10.1 Soldering - Soldering shall be so executed that both a positive electrical and a strong mechanical connection is assured. Joints shall be mechanically secure before soldering and shall not depend on soft solder alone for mechanical strength. Soldering shall meet the requirements of specification MIL-S-6872.

3.4.11 Insulating Sleeving - Insulating sleeving shall be used on the soldered connections to the socket, on all connections to the base of the relay coil and all soldered connections to the base of the contacts.

3.4.12 Standard Tools - The unit shall be so designed that standard tools and test equipment may be used to the greatest possible extent to accomplish all necessary adjustments, maintenance and tests.

3.4.13 Nameplate - A nameplate conforming to the requirements of MIL-STD-130 shall be permanently attached to the top of the cover. The nameplate shall contain the following applicable information:

Relay Bank: Bomb Salvo Circuit
3 Unit, 28 Volt D.C., Type A-1
Federal Stock Number
Contract Number
Serial Number
Manufacturers Part Number
Manufacturers Name

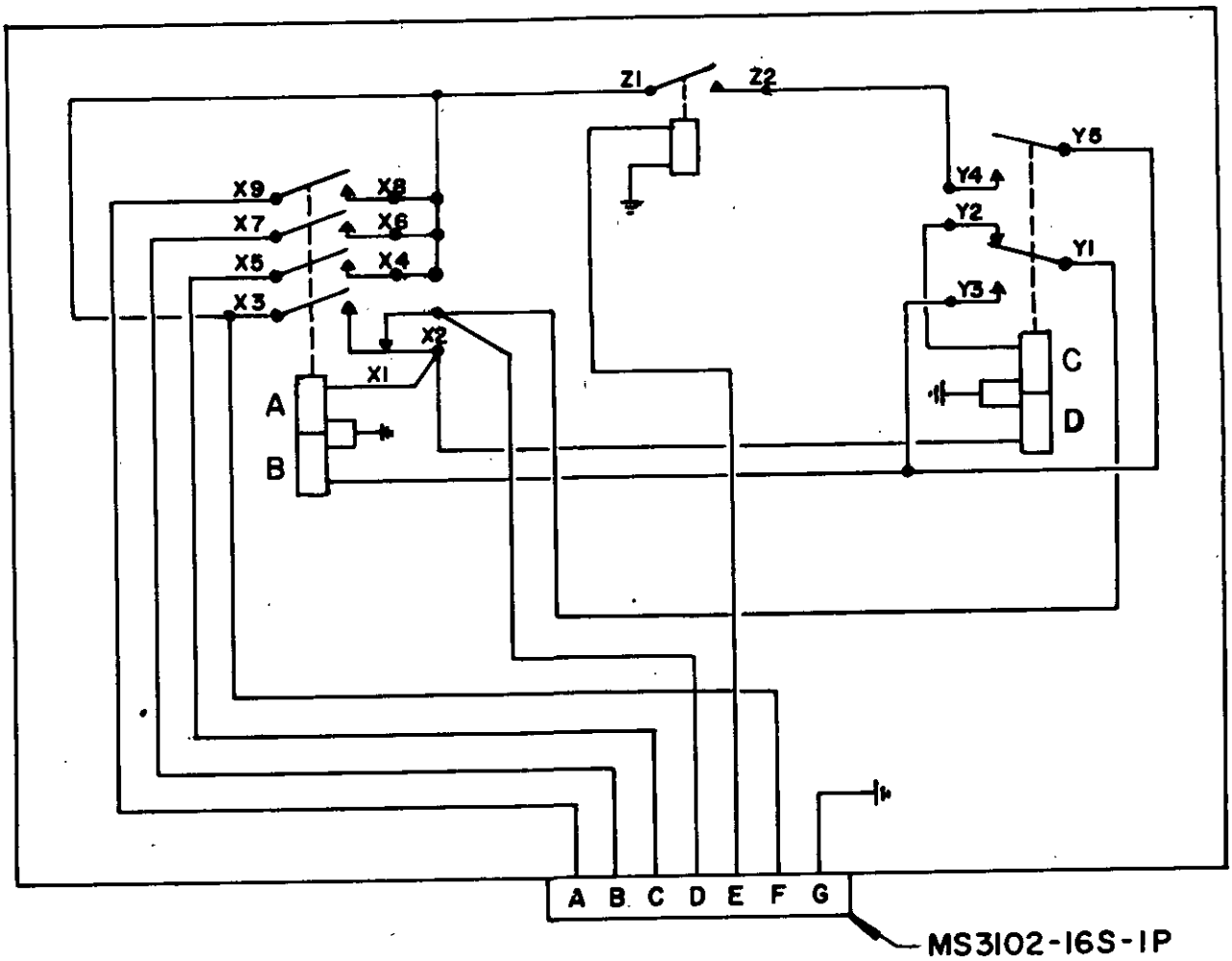
3.4.14 Wiring Diagram - Each item shall be provided with a permanent and legible wiring diagram as shown on Figure 2 and shall be securely attached to or printed on the inside of the box cover.

3.5 Interchangeability - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-D-1000.

3.6 Performance - The relay bank shall meet the performance requirements of this specification when tested as specified herein.

3.7 Workmanship - Workmanship shall be in accordance with highgrade manufacturing practice covering this class of equipment.

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TERMINAL	CONNECTION	CURRENT
A	OUTPUT	3 AMP CAPACITY
B	OUTPUT	3 AMP CAPACITY
C	OUTPUT	3 AMP CAPACITY
D	ON-OFF SALVO CONTROL	.45AMP DRAIN MAX
E	OFF SALVO CONTROL	.04AMP DRAIN MAX
F	SALVO POWER SOURCE	TOTAL OF A-B-C
G	GROUND	.45AMP DRAIN MAX

FIGURE 2 Schematic

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the government. The government reserves the right to perform any of the inspections set forth in the specification when such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Inspection records of the examinations and tests shall be kept complete and available to the Government for a minimum of two years.

4.1.1 Test Equipment & Inspection Facilities - Test equipment and inspection facilities shall be of sufficient accuracy, quality and quantity to permit performance of the required inspections. The supplier shall establish calibration of inspection equipment to the satisfaction of the government. Calibration of the standards which control the accuracy of the inspection equipment shall comply with the requirements of MIL-C-45662.

4.1.2 Additional Inspections - Inspections not specified herein shall not preclude the supplier from taking such additional samples and making such additional inspections as he may deem necessary or desirable to assure conformance of the relay banks to this specification.

4.1.3 Quality Control Procedures - The supplier shall establish a quality control system which complies with the requirements of MIL-Q-9858. Evidence of such compliance shall be submitted to the qualifying activity of this specification as a prerequisite for qualification.

4.1.4 Government - Acceptance of the relay banks shall be based upon verification by the government of the supplier's compliance with the requirements of this specification. The government may, at its option, repeat any or all of the inspections specified herein (see 3.2).

4.1.5 Classification of Tests - This specification requires both preproduction and acceptance testing of the relay banks. The preproduction tests are intended to prove that the design, construction, materials, and workmanship used produce a relay bank that meets all of the requirements of this specification. The acceptance tests are intended to prove that all relay banks supplied under this specification do not in any way differ from those that passed the preproduction tests. The successful completion of these tests does not relieve the vendor of the responsibility for complete compliance with all of the requirements of this specification.

4.2. Preproduction Tests - Preproduction tests shall be performed on samples that have been made using the same production equipment, methods, materials, and type of personnel as the relay banks that are to be supplied as production items, i.e., the test samples shall not be "handmade" by engineers or technicians in a

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laboratory if the production items are made by personnel with a lower skill level in factory conditions. A preproduction test report, as specified in the contract, shall be furnished to show that the relay bank and components meet or exceed the requirements of this specification. This inspection establishes the minimum requirements for the examination and testing of relay banks procured by the government in which case the detail performance criteria is embodied in the text of a contract, specified on manufacturer's drawings or other documents.

4.2.1 Test Specimens - Preproduction test specimens shall consist of four relay banks minimum.

4.2.2 Test Conditions - Test Conditions shall be per MIL-E-5272 and this specification.

4.2.3 Test Description - The preproduction tests shall consist of the following:

Examination of Product
Operating Characteristics
Dielectric
Time Delay
Voltage and Temperature
Vibration
Acceleration

4.2.4 Failure and Retest - Upon failure to pass any of the preproduction tests, testing shall be discontinued and measures taken to correct the cause of failure. Final point of retest shall be determined by the government representative.

4.3 Acceptance Testing -

4.3.1 Vendor Acceptance Tests - The procedure for conducting acceptance tests of the unit shall be prepared by the supplier and shall be subject to approval by the government. Acceptance tests shall be conducted at the supplier's plant subject to the approval of a government inspector.

4.4 Test Conditions -

4.4.1 Tolerances - Tolerances shall be as specified in MIL-E-5272 and as specified in this specification.

4.4.2 Temperature - Preproduction testing shall be at 25°C ± 5°C unless otherwise specified.

4.4.3 Attitude - Unless otherwise specified, the qualification testing shall be conducted with the relay bank mounted in the position most likely to cause malfunctioning.

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4.4.4 Altitude - The sealing and operation of the relay bank shall not be adversely affected by changes in altitude ranging from sea level to 50,000 feet in accordance with MIL-E-5272.

4.5 Test Methods -

4.5.1 Preliminary and Mechanical Tests -

4.5.1.1 Examination of Product - Relay banks shall be inspected to verify that the materials, design, construction, physical dimension, weight, item identification, and workmanship are in accordance with this specification.

4.5.1.2 Sealing - Sealing shall be accomplished after drying out the unit, with relays installed, for two hours at +75°C by use of General Electric Company standard red, Glyptal Catalog No. 1201 or equivalent. The box cover shall be secured by the use of machine screws, removal of which, will permit breaking of the seal and opening the box for maintenance purposes when necessary (see 3.4.6.1).

4.5.2 Performance Tests -

4.5.2.1 Dielectric -

4.5.2.1.1 Dielectric Strength - There shall be no breakdown of insulation when the test voltage is applied between the following points for one minute: between each terminal and all other terminals which do not form a circuit, with unit both energized and de-energized, between all terminals connected together and the case. The test voltage shall be 750 volts rms for 28 VDC system. Any breakdown, arcing, or steady increase of fluctuation in leakage current shall constitute failure. Repeated performance of applicable dielectric test shall not cause failure.

4.5.2.1.2 Insulation Resistance - The insulation resistance of the component shall be measured between all mutually insulated parts at a potential of 100 to 500 volts DC by means such as megohm bridge. This shall be done with the unit in the energized and de-energized position. The resistance shall not be less than 100 megohms.

4.5.2.2 Operating Characteristics - Tests covered in paragraphs 4.5.2.2.1 and 4.5.2.2.2 shall be performed 10 times.

4.5.2.2.1 Timing Cycle - The relay bank shall be tested to determine that the timing cycle is within limits specified while operating within the operating voltage range specified in paragraph 3.4.2. Voltage regulation of the unit shall function correctly when operated at all voltages.

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4.5.2.2.2 Time Delay - The relay bank shall be tested to determine that the time delay is not less than 0.06 of a second over the ranges of voltages and temperatures specified herein.

4.5.3 Environmental Tests -

4.5.3.1 Vibration - The relay bank shall be subjected to vibration with an amplitude between 0.003 and 0.005 of an inch at frequencies of 500 to 3000 cycles per minute to determine that the natural frequency of the relays or other component parts do not occur in the frequency range. The unit shall be vibrated along each of the three mutually perpendicular axis which are orientated from the position on which the unit is most susceptible to vibrating motion. The frequency scan shall be conducted slowly and carefully to determine resonate frequencies. The frequency range shall be broken into small convenient intervals and each interval scanned at a constant applied acceleration and amplitude. After the frequency scan, the unit shall be subjected to a resonance endurance test for eight hours at each resonate frequency.

4.5.3.2 Vibration with Power - The relay bank shall be continuously connected to a power source while being vibrated for a three hour period at various frequencies between 1000 and 2400 cycles per minute with an amplitude of from 0.018 to 0.020 of an inch, after the vibration test specified in paragraph 4.5.3.1. During this test, the relay bank shall be operated through 12 complete cycles of operation each hour, by applying impulses of not less than 0.10 seconds duration. Each complete cycle shall comprise:

a. 1st impulse to terminal D, relay bank functions.

b. 2nd impulse to terminal D, relay bank returns to normal.

c. 3rd impulse to terminal D, relay bank functions.

d. 4th impulse to terminal E, relay bank returns to normal.

During this test, contacts that should be open shall not close and contacts that should be closed shall not open, due to vibration. No looseness in the relays or damage to any part shall result from the three-hour vibration period.

4.5.3.3 Acceleration - The relay bank shall be subjected to acceleration forces of 10 gravity units (10g's), after being subjected to vibration test. The time required to obtain the 10g acceleration shall not be more than two seconds. The relay bank shall be mounted in the position or positions most likely to cause malfunctioning and shall be accelerated for a minimum of one minute. During this test, the relay bank shall function satisfactorily through one cycle of operation as specified in paragraph 4.5.3.2, also contacts that should be open shall not close and contacts that should be closed shall not open due to application of specified acceleration forces.

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4.5.3.4 Temperature Stabilization - Temperature stabilization shall be considered to be accomplished when, with a constant temperature outside the case, the temperature inside the case has remained constant for thirty minutes. The time required to reach stabilization may be determined experimentally for the first item on any contract and under given test conditions and this time used on subsequent tests under the same conditions to permit testing without breaking the seal of the banks under test.

5. PREPARATION FOR DELIVERY

5.1 Packaging, Packing and Marking - The packaging, packing, and marking requirements specified herein apply only to direct purchases by or direct shipments to the government.

5.1.2 Interior Packaging - Unless otherwise specified, each relay bank shall be packaged in accordance with MIL-P-6064 except that the metal container shall be in accordance with MIL-D-6055.

5.1.3 Exterior Packaging - Unless otherwise specified, all items shall be packed for domestic shipment. Each shipping container, insofar as possible, shall contain the identical number of articles, shall be of uniform size, and shall be designed to inclose the contents in a snug, light fitting manner. The gross weight of the exterior shipping container when packed for shipment shall not exceed approximately 200 pounds.

5.1.4 Domestic Shipment - Unless otherwise specified, for domestic shipment, interior packages shall be packed in substantial commercial exterior shipping containers to insure acceptance by common or other carrier for safe transportation, at the lowest rate, to the point of delivery. Except as specified herein, the container shall conform to the requirements of Uniform Freight Classification Rules in effect at the time of shipment. Containers shall be capable of withstanding storage, rehandling, and reshipment without the necessity of repacking.

5.1.5 Export Shipment - Unless otherwise specified, for export shipment, the interior containers shall be packed in accordance with MIL-D-6055.

5.1.6 Marking Interior - Each interior package shall be durably and legibly marked with information as specified in paragraph 3.4.13.

5.1.7 Marking Exterior - Each exterior shipping container shall be marked in accordance with MIL-STD-129.

6. NOTICE

When government drawings, specifications or other data are used for any purpose other than in connection with a definitely relative government procurement.

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operation, the United States Government thereby incurs no responsibility whatsoever and the fact that the government may have formulated, furnished or in any way supplied the said drawings, specifications, or other data is not to be regarded by implications or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
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Preparing activity:
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